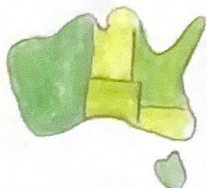
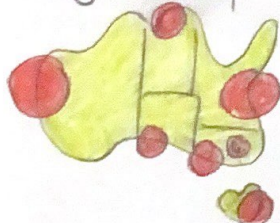


IDEAS

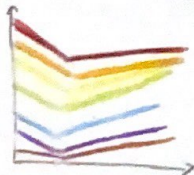
① choropleth map



② symbol map



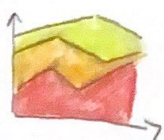
③ multi-line chart



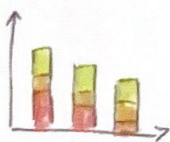
④ Grouped bar chart



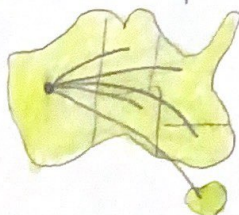
⑤ stacked Area chart



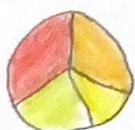
⑥ stacked bar chart



⑦ Flow map



⑧ Pie chart

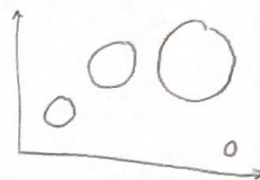


⑨ Heatmap



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⑩ Bubble chart



FILTER

⑤, ⑥, ⑦, ⑨, ⑩ was not considered because:

- ⑤ difficult to compare individual categories accurately
- ⑥ similar reasons to stacked area chart, hard to compare individual categories
- ⑦ The dataset does not include origin - destinate flow data
- ⑨ similar idea with choropleth map
- ⑩ similar idea with symbol map

CATEGORISE

Spatial comparison

- choropleth map
- symbol map

Temporal comparison

- multi-line chart

Categorized comparison

- Grouped bar chart
- pie chart

COMBINE & REFINE

- ① Combine two multi-line chart to compare domestic vs international.
- ② Grouped bar chart show Top 5 countries per year across 2019-2024
- ③ pie chart breakdown travel reasons for selected year.

Questions

- ① Do they clearly show domestic vs international travel trends?
- ② Does the chosen idiom help target audience understand tourism pattern across Australia?
- ③ Does the combination of charts allow users to compare states, year and visitor types effectively?

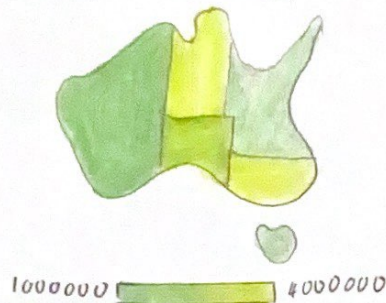
LAYOUT

TOURISM IN AUSTRALIA

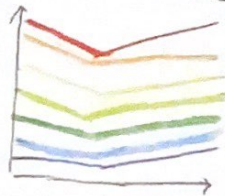
select Year

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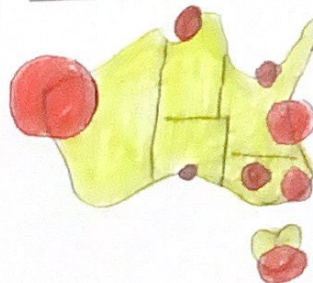
Domestic Tourism



Domestic spendings



International arrivals



International spendings



■ NSW
■ VIC
■ ACT
■ NT
■ QL
■ SA
■ TAS

FOCUS

ESSENCE: Emphasise geographical distribution and trends

key ideas: Users can quickly see which states have higher domestic trips, where international visitor concentrate, and spending pattern over time.

COMPONENTS / OPERATIONS

Choropleth map components

- state boundaries, colour scale, tooltip

Symbol map components

- Locations, circle size, tooltip

Multi-line chart components

- x-axis: Year, y-axis: spend, lines: domestic vs international, legend

Operations:

- Hover to see tooltip
- Click legend to highlight
- Select to filter year

PROS

- Clear spatial comparison between domestic and international tourism
- Interactive element (tooltip, legend, year slider) improve exploration

CONS

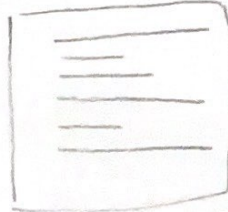
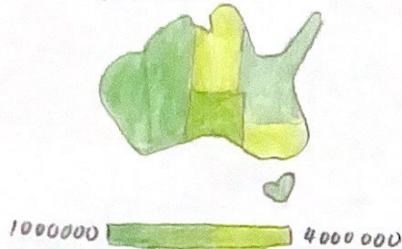
- Only shows main pattern, less insight on top 5 visitor countries or main travel reason
- might be overwhelming if all interactivity is used at once.

LAYOUT

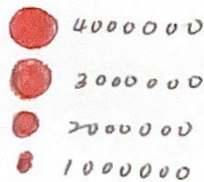
TOURISM IN AUSTRALIA

select year

Domestic Tourism



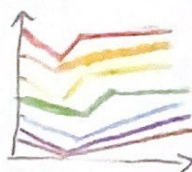
International arrivals



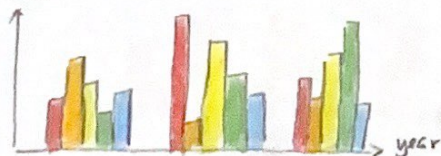
Domestic spendings



International spending



spends TOP 5 Countries



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FOCUS

Essence:

- Provide a comprehensive view of tourism trends
- Audience see patterns geographically, financially and categorical

OPERATIONS

- ① Hover on map, grouped bar charts and line chart
 - show tooltips
- ② Click on state or country in legend
 - highlight corresponding lines on bar.
- ③ Year selection:
 - All visualisation update dynamically to reflect selected year.

PROS

- Multi-dimensional view allow quick understanding of tourism trends across geography, spending, and visitor profiles.
- Interactive features let the client explore specific states, countries or years.
- Adding bar chart provides complementary perspectives.

CONS

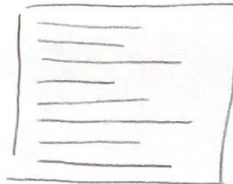
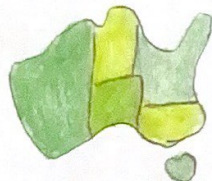
- users may need guidance to understand correlations across multiple charts
- Requires clean & consistent colour schemes to avoid confusion between maps and charts.
- Adding interactivity increases implementations complexity in the final dashboard.

LAYOUT

TOURISM IN AUSTRALIA

Select Year

Domestic Tourism

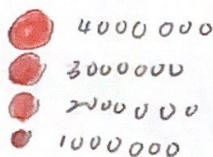
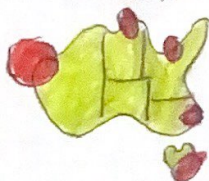


1000000

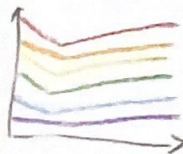


4000000

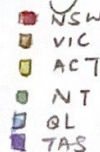
International Arrivals



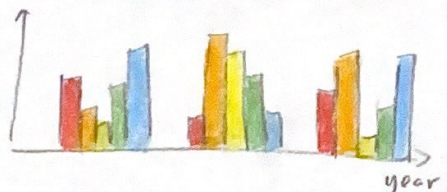
Domestic spendings



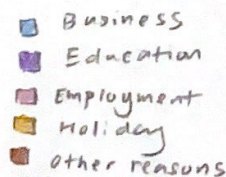
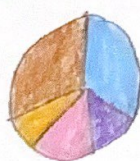
International spending



TOP 5 Countries



Main Reasons to travel



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ID = 34070397

date : 13/10/2025

sheet : 4

FOCUS

Essense :

- Main idea is still interactive exploration, but now includes categorical analysis of travel reasons
- Audience can see which reasons dominate travel.

OPERATIONS

- 1) Hover :
show tooltips for maps, bar charts and pie charts
- 2) Click legend items :
Highlight state or country across all charts.
- 3) Drag year slider
updates all visualisations to the chosen year.

PROS

- Pie chart add categorical insight to the temporal & spatial views
- Interactive exploration allows deeper analysis by reason of travel

CONS

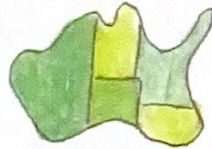
- Pie charts may hide small or be harder to read with too many categories
- Adding more charts increases screen complexity, so layout must be carefully considered.

LAYOUT

TOURISM IN AUSTRALIA

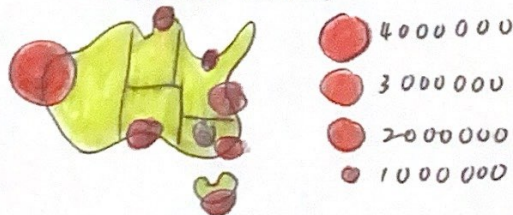
select year — 2024

Domestic Tourism

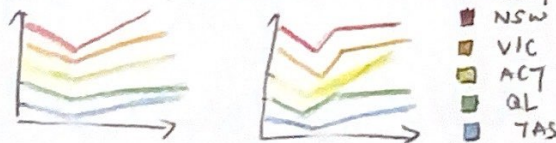


1000000 4000000

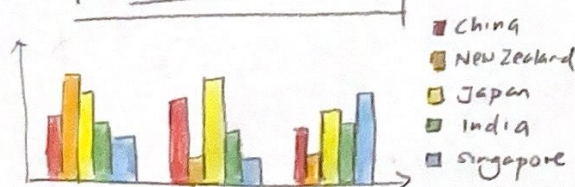
International Arrivals



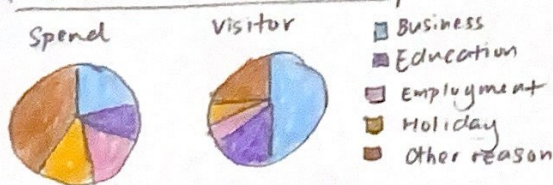
Domestic vs International Spendings



TOP 5 Countries



Main Reason to travel



Name: mm Yee Wong
ID: 34070397

Date: 13/10/2025

Sheet - 5

FOCUS

ESSENCE:

- A top-to-bottom, layered dashboard showing spatial, temporal and categorical insight

COMPONENTS

- ① Choropleth map
 - raw total trips per state
- ② symbol map
 - International visitor arrivals
- ③ dual multi-line chart
 - domestic vs international spending trends
- ④ grouped bar chart
 - top 5 countries
- ⑤ pie charts
 - trip vs spending by reason travelling
- ⑥ Year slider
- ⑦ Legends & tooltips.

OPERATIONS

- ① Year sliders
 - updates all charts dynamically
- ② Hover
 - show tooltips
- ③ Legend selection
 - highlights corresponding entities across charts

DETAILS

Libraries / Tools:

- Vega-Lite
- TopoJson
- CSV

Algorithm & Data handling

- ① Choropleth map
 - colour scaled using raw total trips
- ② symbol map
 - proportional sizing using visitor count

Development Notes:

- ① Ensure fixed colour scales for maps & charts
- ② Horizontal concatenation for line charts & pie charts
- ③ Annotation tooltips for major events.